TIKA 4431 MIME detection with number signs

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This contribution fixes a filename-hint error in Apache Tika core. CSV content named Lorem-Ipsum#123.csv was classified as text/plain because URI parsing removed the part containing the .csv extension. The change preserves that extension for filename-like inputs while retaining existing handling of explicit URLs. The published group patch has regression coverage and recorded peer review. [1-4]

Problem and relevant code

The original report compares identical content under Lorem-Ipsum.csv and Lorem-Ipsum#123.csv. The first name yields text/csv; adding the number sign changes the result to text/plain. The reproduction uses UTF-8 bytes for a short semicolon-separated document, so the input content stays constant while the resource name changes. [1,4]

```
Lorem;Ipsum;
1;2;
3;4;
```

Tika.detect(byte[], String) passes the document name into the stream-based API, which stores it as TikaCoreProperties.RESOURCE_NAME_KEY. Within the detection process, MimeTypes reads content evidence and then uses the resource-name hint to refine compatible candidate types. The affected branch constructs a java.net.URI and extracts its path before looking up the filename. [3,9]

A bare name such as Lorem-Ipsum#123.csv is also valid relative-URI syntax. URI parsing interprets #123.csv as a fragment, leaving the path Lorem-Ipsum. Consequently, the filename lookup loses the .csv hint that can specialise the sample content from plain text to CSV. The fix belongs in this name-extraction step; the CSV input and MIME registry do not need changes. [1,3]

Implementation and compatibility

The patch adds ten production lines to MimeTypes.java. After URI construction, the additional rule runs only when both the scheme and raw authority are absent. It locates the first question mark, then compares the final raw number sign and dot before that boundary. If a number sign precedes the last dot, literal number signs are escaped as %23 and the URI is reparsed. Existing path extraction, filename lookup and hint application then continue. [3]

This condition preserves the filename heuristic without reinterpreting explicit URL fragments. It uses original characters before percent decoding, so %2E does not become a dot for the boundary decision. Restricting the comparison to the part before the first question mark also prevents a dot in a query value from driving that decision. [3,4]

Regression coverage and design limits

The 32 new parameterised inputs exercise both content-plus-name detection and name-only detection. Alongside the existing test, JUnit reports 33 detection cases. Coverage includes ordinary and repeated number signs, uppercase extensions, dotted names, relative and absolute paths, Windows paths, spaces, encoded hashes, query strings, explicit URI schemes and network authorities. HTTP and FTP script-name controls preserve their existing context-dependent results. [4]

Resource name	Expected result
report.txt#section.csv	text/csv
https://example.test/report.txt#section.csv	text/plain
report.txt#section%2Ecsv	text/plain
report.csv#section?download=.txt	text/csv

These examples state a deliberate compatibility choice. A scheme-less string can denote either a filename or a relative URI, so the rule cannot determine caller intent in every case. The bare name above supplies a CSV filename hint, while the explicit URL retains its .txt path hint. A filename hint is not proof that arbitrary content is CSV. This ambiguity remains a documented limitation of an API that accepts names, paths and URLs as strings. [2-4,9]

Validation of the published patch

The following results were recorded on 21 October 2026 for the group-base patch, using Windows and JDK 21. The inspected checkout remains at published head 86483a6. On 29 October, SHA-256 checks matched all three changed files to the recorded validation snapshot, and PR 7 still identified the same head. The table cites those dated executions for the published patch. [2,8]

Check	Reported cases	Failures	Errors	Skipped
Detection tests with production fix removed	33	9	0	0
Detection tests with production fix restored	33	0	0	0
Core unit run	1,391	0	0	11
Alternate XML provider run	12	0	0	2
Core integration test TikaIT	1	0	0	0

The negative control retained the new tests and removed only the production fix. Its nine failures demonstrate that the regression cases distinguish the faulty implementation from the corrected one. The restored patch also passed annotation-processor tests, Checkstyle, Spotless, forbidden-API checks and Apache RAT, ending with Maven BUILD SUCCESS. Counts above belong to separate executions and are not added into a single total. [2,8]

```
./mvnw.cmd clean install -pl tika-core -am -Ppedantic -Dtika.test.locale=random
```

The command omits the machine-specific local Maven-cache argument. Validation covers tika-core and reactor prerequisites. It does not establish a successful build of every Tika module, remote CI success, or upstream acceptance. The current group head had no GitHub check runs returned by the inspected API endpoint. No performance benchmark was recorded. [2,8]

Review and contribution process

The change is limited to MimeTypes.java, TikaDetectionTest.java and CHANGES.txt: 77 added lines and no removed lines. No public API or dependency is added. The changelog identifies the user-visible detection correction, and the commits reference TIKA-4431. This follows the project guidance to keep fixes focused, include a regression test that fails without the change, document user-visible changes, and identify the Jira issue. [2,7]

The patch and its evidence were shared in group issue 1 before publication as the group PR. On 17 October, Jiajun Li reviewed the production logic, all 32 new parameterised inputs and the changelog. His isolated Java check passed 44 cases; he did not independently rerun the full Tika suite. On 21 October, Jintian Zhang reviewed the same areas and recorded an AI-assisted isolated check passing 17 parsing cases. Both reviews reported no blocking findings and asked for the bare-filename versus explicit-URL distinction to remain clear in an upstream PR. [5,6,10]

The reviewed additions were ported to group base 76b33fa while preserving existing group changes. The final three commits are 75dbff0 for the changelog, 4a76680 for the production change, and 86483a6 for the regression tests. Group PR 7 targets the group fork. As checked on 29 October 2026, it was open and unmerged. An Apache PR search for TIKA-4431 returned no matching pull requests. Upstream submission and maintainer acceptance are therefore not claimed here. [2,11]

Individual contribution and collaboration

I am the accountable owner of TIKA-4431 under u8026876. My contribution comprises the filename-hint correction, regression coverage and changelog in the three commits listed above. I used Codex to assist the investigation, implementation, test design and execution, porting, publication preparation, and explanation. The commits carry Generated-by: Codex trailers, and the issue and PR disclose this assistance. [2,10]

Jiajun Li (Lynnbb88) and Jintian Zhang (zjt060523terry) contributed scoped peer reviews and the isolated logic checks described above. Their review evidence is separate from the core Maven validation. I also used Codex to help prepare this report section from the cited records and my account of the main challenge. [5,6,10]

Reflection and possible improvements

My main challenge was deciding what correct behaviour should mean when the input could follow either URL or filesystem conventions. In a filename, a number sign can be literal; in a URI, it begins a fragment. Preserving everything after it could fix the reported filename while changing explicit URL behaviour. I had to weigh these competing conventions and make a reasoned decision about which interpretation the fix should preserve.

The final choice retains URI handling when parsing identifies an explicit scheme or authority. For inputs without either, it applies the narrower filename heuristic described above. This uses structural information and a documented rule; it cannot prove caller intent. The lesson for me was that correctness in this case required making an ambiguous input contract explicit and checking the chosen boundary with tests. [3,4,9]

The investigation also exposed two implementation traps: decoded fragments made encoded dots look like extension boundaries, while a later candidate still let a query-string dot affect the decision. Saved records show two initial-candidate failures and one intermediate query-related failure. The final raw-character comparison excludes both cases. [8]

A future API that distinguishes filenames, paths and URLs explicitly could reduce this ambiguity, subject to compatibility work. Broader integration validation and measured performance checks remain possible follow-ups. [3,4,8,9]

References and evidence

Source links below identify the issue, the exact published source revision and recorded peer reviews. Live PR status and the bounded upstream search were checked on 29 October 2026. Validation runs cited in the section were performed on 21 October, with earlier candidate investigations on 15 October.

[1] TIKA-4431 original bug report.
https://issues.apache.org/jira/browse/TIKA-4431

[2] Group PR 7 implementation validation and submission status.
https://github.com/u8026876/comp2120-gyg-a5/pull/7

[3] MimeTypes source at the published commit.
https://github.com/u8026876/comp2120-gyg-a5/blob/86483a61d15e22a03f1bbdf6e0c199012baa34ed/tika-core/src/main/java/org/apache/tika/mime/MimeTypes.java#L564-L611

[4] TikaDetectionTest regression inputs at the published commit.
https://github.com/u8026876/comp2120-gyg-a5/blob/86483a61d15e22a03f1bbdf6e0c199012baa34ed/tika-core/src/test/java/org/apache/tika/TikaDetectionTest.java#L31-L95

[5] Jiajun Li scoped peer review.
https://github.com/u8026876/comp2120-gyg-a5/issues/1#issuecomment-6233761379

[6] Jintian Zhang scoped peer review.
https://github.com/u8026876/comp2120-gyg-a5/issues/1#issuecomment-6311411395

[7] Apache Tika contribution guidance.
https://tika.apache.org/contribute.html

[8] Local validation evidence collected on 21 October 2026. See the attached evidence folder: verification-summary.json, core-clean-install.log, and the two detection XML reports. Earlier candidate logs support the iteration account.

[9] Tika detection API source at the published commit.
https://github.com/u8026876/comp2120-gyg-a5/blob/86483a61d15e22a03f1bbdf6e0c199012baa34ed/tika-core/src/main/java/org/apache/tika/Tika.java#L183-L334

[10] Group issue 1 ownership and review record.
https://github.com/u8026876/comp2120-gyg-a5/issues/1

[11] Apache Tika ticket search checked on 29 October 2026.
https://github.com/apache/tika/pulls?q=is%3Apr+TIKA-4431

Reproducibility record

Published group head: 86483a61d15e22a03f1bbdf6e0c199012baa34ed. Group base: 76b33fa0ea016ceb50b7941b8176252855e8ac4d. The attached evidence folder includes the production-fix negative control, the restored detection report, the core build log, file hashes and a machine-readable verification summary. The earlier candidate logs are included to support the design-iteration account.